

Arenic Authoring and Recording Runbook

A practical onboarding guide for designers building Normal, Heroic, and Mythic boss encounters with the native authoring tool.

AUDIENCE

Encounter designers and content authors

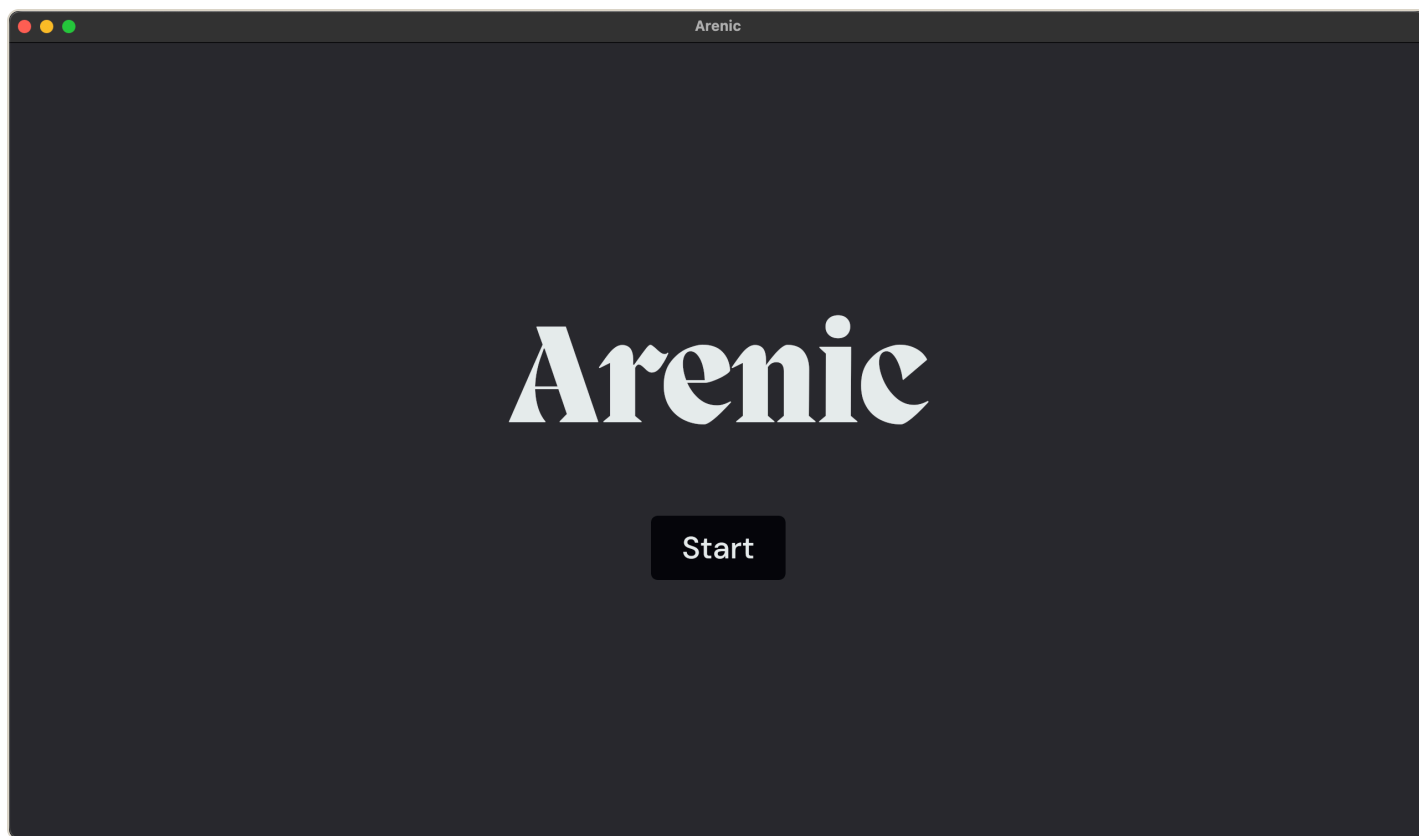
TOOL

Native author build:
`just author`

OUTPUT

Versioned encounter stacks in
`assets/encounters`

Current ability caveat. Not all boss and hero abilities exist yet. Today, boss slots can record, but the live loadout is still placeholder-heavy: slot 1 resolves to Holy Nova and slots 2-4 may be inert until their abilities land. Build the timing, movement, tile pressure, and layer structure now; swap in final abilities later.

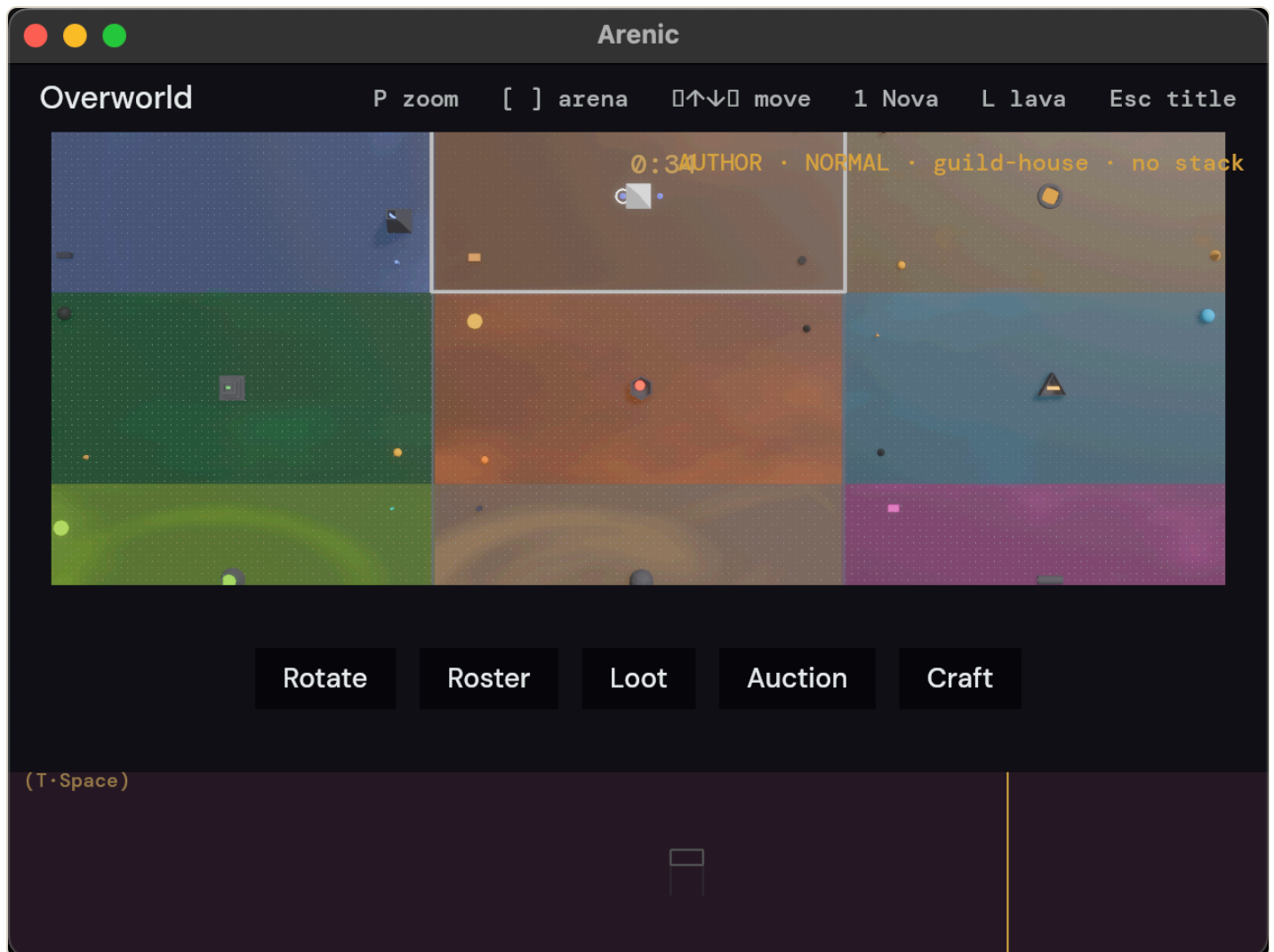


Start from the native app. The web build does not include author mode.

1. Mental Model

Author mode is a native-only build that layers encounter content on top of the normal game simulation. Each arena and difficulty owns a draft layer stack. Playback folds that stack into the normal two-minute, 60 Hz timeline.

- **Cycle length:** 120 seconds, 7,200 ticks. Every movement, ability, tile keyframe, and effect key lands on an integer tick.
- **Difficulty ownership:** Normal, Heroic, and Mythic are separate authoring targets. Use `D` to cycle between them.
- **Draft vs published:** edits stay in memory until `W` publishes the focused arena's stack.
- **Layer conflict rule:** top tile layer wins when two layers claim the same cell at the same tick.
- **Rollback:** delete the newest `layers.vNNNN.ron` file, then restart/reload. Git history is the second level of undo.

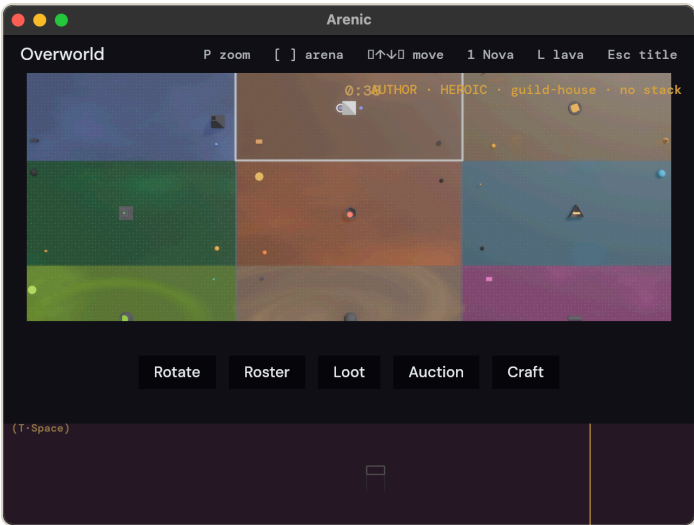


The author chip shows difficulty, arena slug, stack state, layer count, and dirty state.

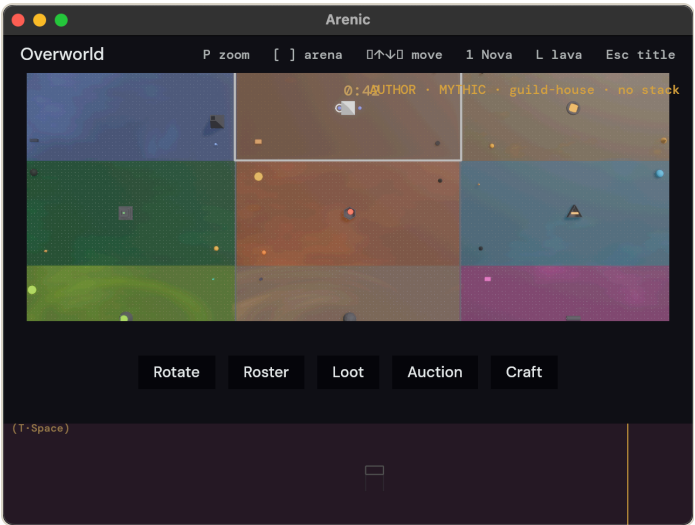
2. Launch and Pick the Assignment

1. Run `just author`. Do not use the web app for authoring; author systems are excluded from WASM builds.

2. Use **[** / **]** to focus the arena you are building.
3. Use **D** to choose Normal, Heroic, or Mythic. If the tool warns about dirty edits, publish with **W** or quit/reopen to discard the in-memory draft.
4. Use **P** to zoom into the focused arena for detailed placement.



Heroic target selected.



Mythic target selected.

3. Global Hotkeys

Input	Use
<code>P</code>	Toggle overworld and single-arena zoom.
<code>[/]</code>	Cycle focused arena. In zoomed view, the camera follows.
<code>D</code>	Cycle authoring difficulty: Normal -> Heroic -> Mythic.
<code>B</code>	Possess the focused arena's boss, then each minion layer, then return to puck control.
<code>M</code>	Add a token minion layer at the playhead. The entity browser is preferred for deliberate placement.
<code>E</code>	Open the entity browser.
<code>T</code>	Toggle tile editor mode.
<code>W</code>	Publish the focused arena's entire draft stack as the next <code>layers.vNNNN.ron</code> .
<code>F5</code>	Restart the focused arena at tick 0 for a fresh watch-through.
<code>F1</code> or <code>?</code>	Toggle the in-app dope sheet binding overlay.

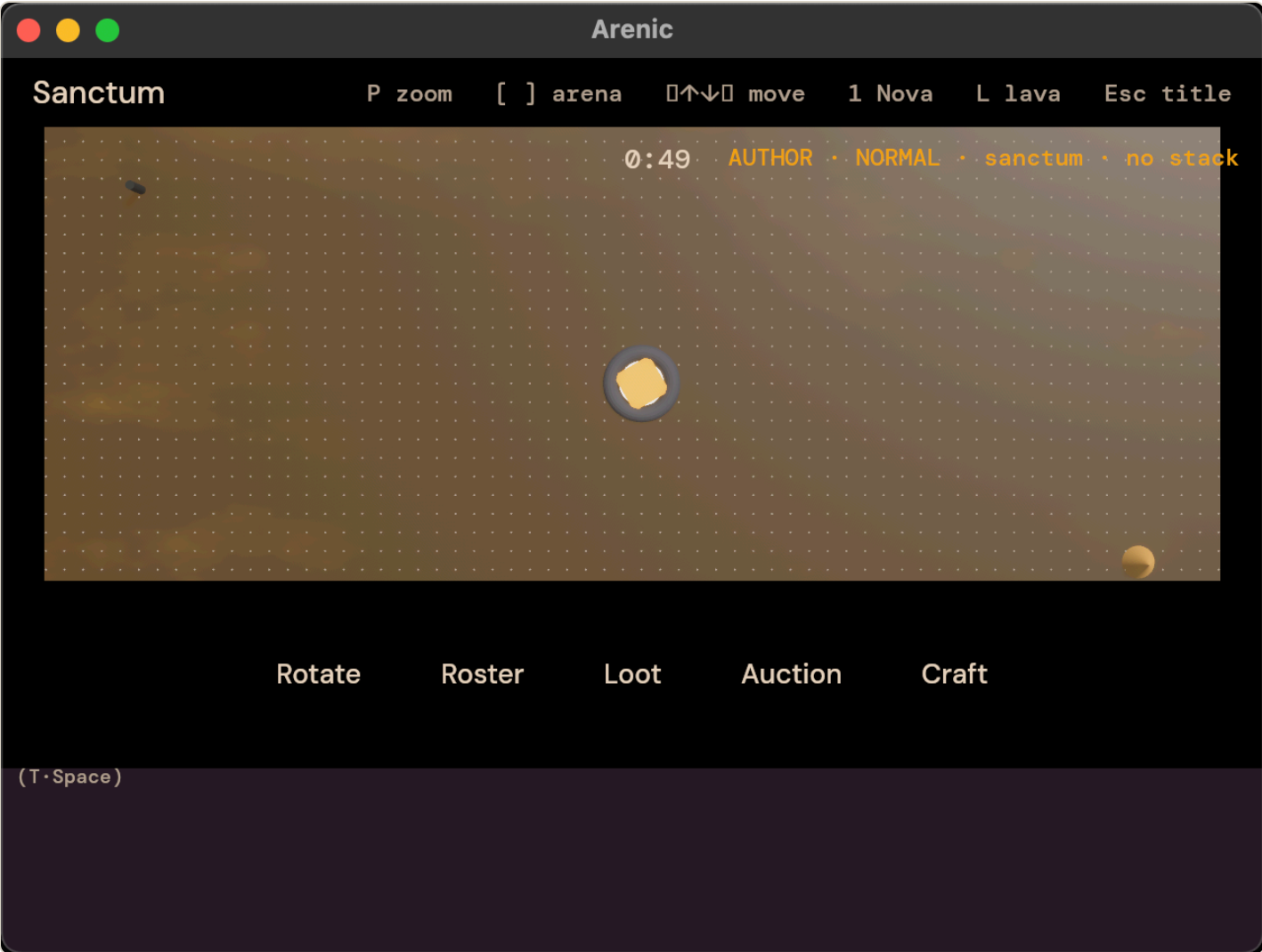
World and Recording Basics

Input	Use
<code>Arrow keys</code>	Move the selected puck, boss, or minion one tile per press. During tile mode, arrows move the tile cursor instead.
<code>1-4</code>	Cast the selected entity's ability slot. For possessed bosses, slots resolve through the boss phase loadout.
<code>R</code>	Start a recording countdown, stop a recording early, or open the relevant recording modal.
<code>Tab</code>	Switch selected hero in normal world control. During tile mode, cycle tile layers.
<code>Esc</code>	Cancel modal / return to title when idle.

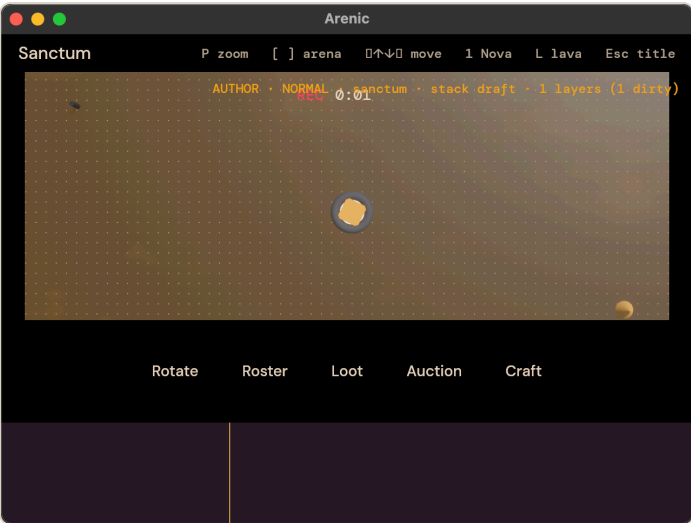
4. Record Boss and Minion Timelines

1. Focus a combat arena with `[ / ]`, then zoom with `P`.
2. Press `B` until the boss or target minion is selected.
3. Rehearse movement with arrows and slot casts with `1-4`. Hold an arrow while pressing a slot to record an aim direction for abilities that use it later.
4. Press `R` to start the three-second countdown. Capture runs from 0:00 across the two-minute cycle.

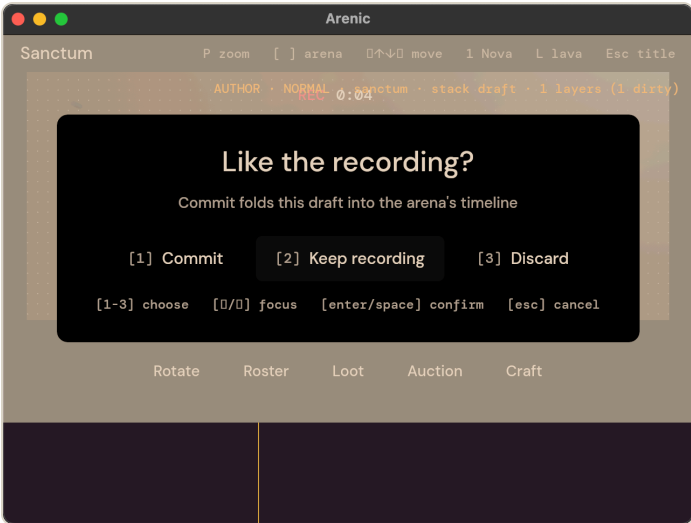
- 5. Press **R** again to stop early, then choose **Commit**, **Keep recording**, or **Discard**.
- 6. Only publish with **W** when the take should become the next versioned encounter file.



Possess a boss or minion before recording. Bosses and minions use the same record/replay flow as heroes.



REC means the current staff is capturing input intent.

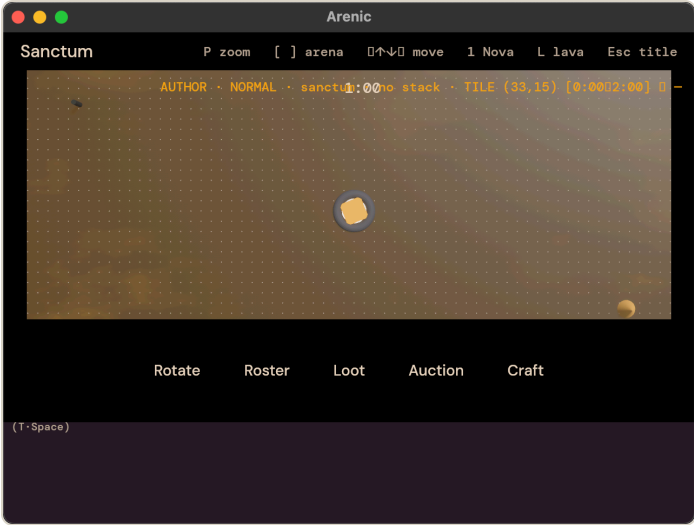


Use number keys to resolve recording modals quickly.

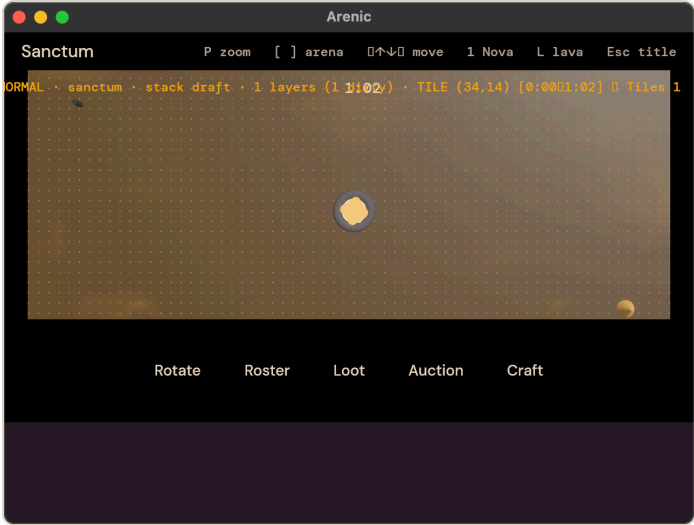
5. Build Tile Choreography

Tile authoring is for floor hazards, safe windows, waves, and phase pressure. The editor writes tile keyframes into the selected tile layer of the draft stack.

Input	Tile editor behavior
T	Open/close tile mode. The focused arena pauses while editing.
Arrow keys	Move the tile cursor one cell.
, / .	Scrub playhead back/forward one second. Hold Shift for ten seconds.
I / O	Set in/out marks for the keyframe range.
Space	Toggle a Lava keyframe for the cursor cell over the marked range.
N	Add a new tile layer on top of the stack and select it.
Tab	Cycle the paint target through existing tile layers.



Tile mode exposes cursor cell, mark range, and target layer in the author chip.

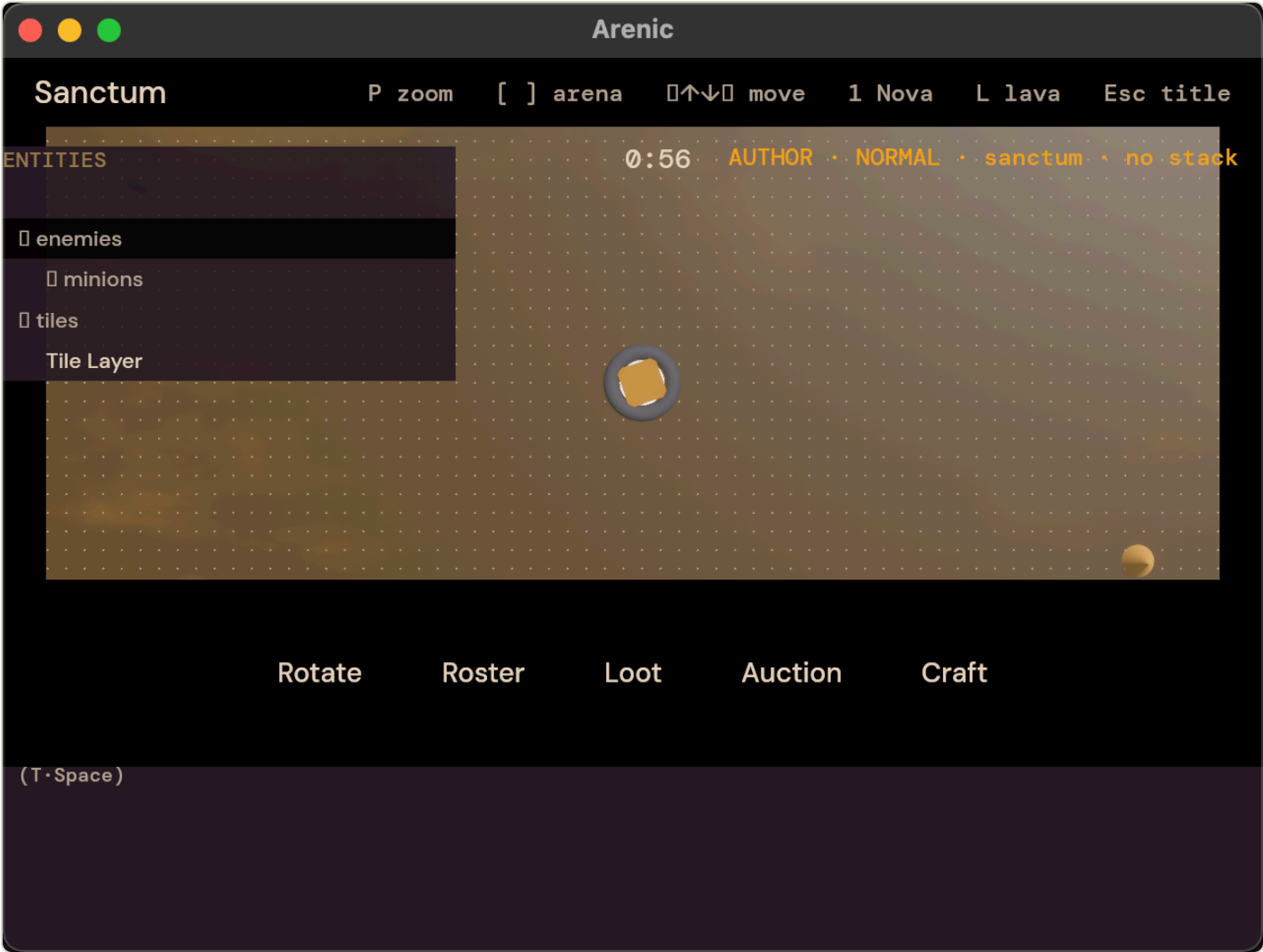


After painting, the stack becomes dirty until published.

6. Use Layers, Browser, and Dope Sheet

Entity Browser

Input	Use
E	Open/focus the browser.
Up / Down	Move focused row.
Right / Left	Expand/collapse directories or jump to parent.
Type	Filter by path substring. Backspace edits the filter.
Enter	Add focused palette entry at the playhead.
Esc	Close filter first, then browser.

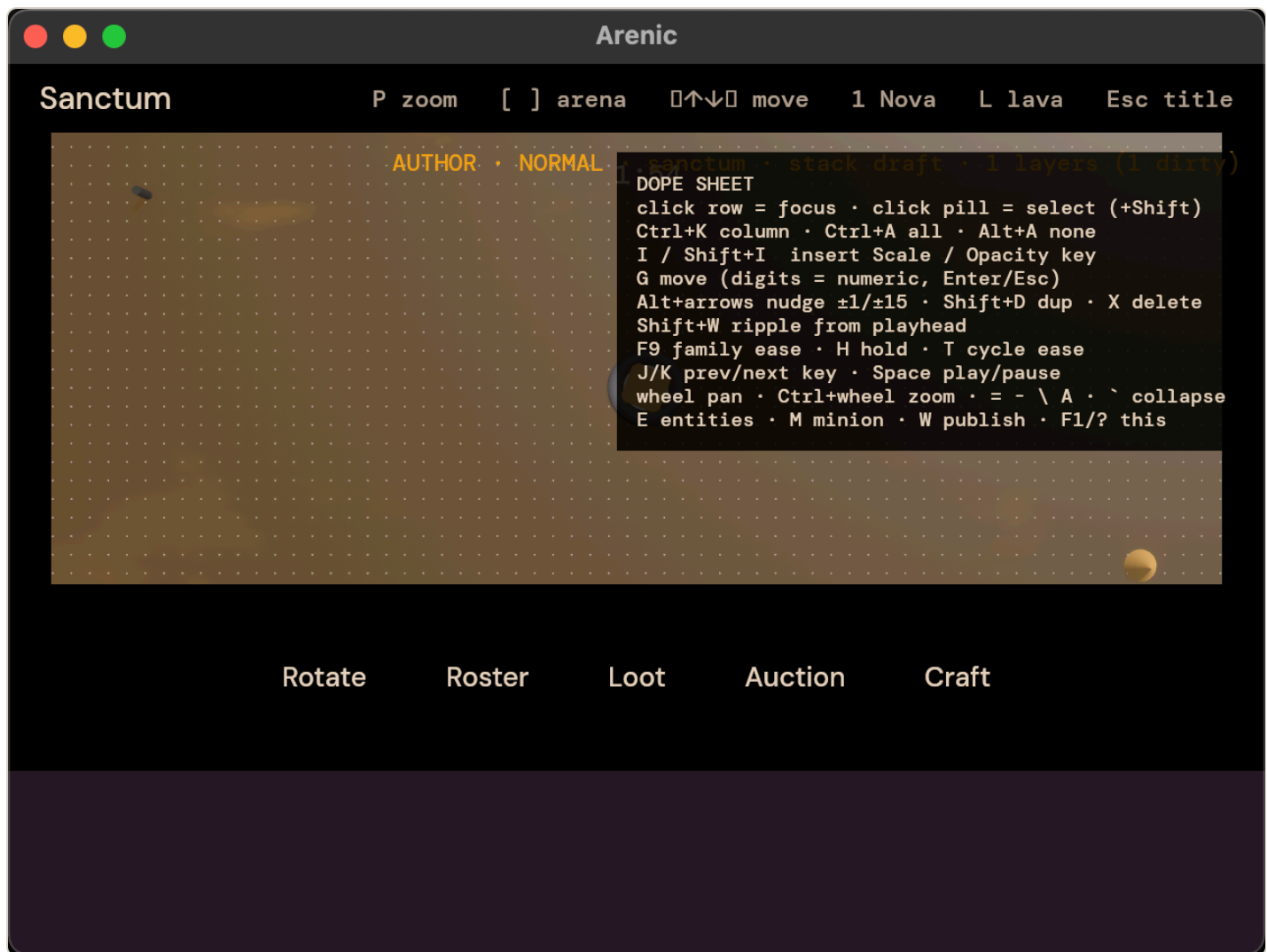


The browser currently exposes Token Minion and Tile Layer entries. More archetypes can be added as palette data grows.

Dope Sheet and Effects

The dope sheet edits effect tracks on layers. It is also the fastest way to inspect row focus, playhead position, and the current key-editing bindings.

Input	Use
Click row / pill	Focus layer row or select an effect key. Shift-click adds to selection.
Ctrl+K	Select keys at the playhead column.
Ctrl+A / Alt+A	Select all / clear selection.
I / Shift+I	Insert Scale / Opacity effect key at the playhead.
G	Move selected keys. Digits enter a numeric tick delta; Enter commits; Esc cancels.
Alt+Left/Right	Nudge selected keys by one tick; add Shift for 15 ticks.
Shift+D	Duplicate selected keys into move mode.
X / Delete	Delete selected keys.
Shift+W	Ripple keys at or after the playhead.
F9 family / H / T	Set ease-in-out, hold, or cycle the selected key's ease.
J / K	Jump playhead to previous/next key.
Wheel / Ctrl+Wheel	Pan / zoom the timeline.
=, -, \, A, `	Zoom in/out, toggle full cycle, frame all, collapse panel.



Use **F1** in-app for the live binding overlay. Some design-doc target interactions are intentionally deferred; this overlay reflects the implemented surface.

7. Designer Assignment: Build Three Difficulties

Each assigned boss should receive a Normal, Heroic, and Mythic pass. Because final abilities are incomplete, focus on readable timing, repeatable movement, tile pressure, and layer structure.

Normal

- Create a readable two-minute boss movement loop with clear safe windows.
- Use slot 1 where you need a placeholder cast. Do not depend on missing slots 2-4 for shipped difficulty.
- Add at least one simple tile hazard or phase marker if it teaches the encounter pattern.
- Publish only after watching a full clean cycle with `F5`.

Heroic

- Start from the Normal mental model, then add one additional pressure layer: tighter movement, a tile layer, or a minion layer.
- Keep readability: designers and players should still be able to explain why a failure happened.
- Use the same two-minute phase boundaries so later ability swaps remain aligned.

Mythic

- Increase density through layering, not random input spam. The score should still be deterministic and inspectable.
- Use separate tile layers for different concepts so top-layer conflict wins are intentional.
- If adding minions, record each staff separately and verify the folded result from tick 0.

Done Criteria for Each Difficulty

- The author chip has no dirty draft after publish.
- The latest file exists under `assets/encounters/<arena>/<difficulty>/layers.vNNNN.ron`.
- A full `F5` watch-through starts correctly from tick 0.
- Any placeholder ability expectations are noted in the encounter handoff.

8. Troubleshooting

Symptom	Fix
Author HUD missing	Use <code>just author</code> . The web app and normal game build intentionally exclude author mode.
<code>D</code> will not switch difficulty	You have dirty draft edits. Publish with <code>W</code> or quit/reopen if the draft should be discarded.
Published the wrong take	Delete the newest <code>layers.vNNNN.ron</code> in that arena/difficulty directory, then restart/reload. Do not hand-edit another difficulty's folder.
Ability slot does nothing	Expected for unfinished abilities. Record the timing anyway only if the future slot needs that timestamp.
Modal blocking inputs	Use <code>1-9</code> , focus arrows/Tab, Enter/Space, or Esc. World input is gated while modals are open.
Tile layer conflicts look wrong	Remember top layer wins. Put overrides above broad base patterns.